

MAT1002 Lecture 5, Monday, Jan/25/2021

Outline

- Power series (10.7)
 - ↳ Radius of convergence
 - ↳ Operations
- Taylor series (10.8)
- Taylor's theorem (10.9)

Radius of Convergence

Theorem (Existence of the Radius of Convergence)

For any power series $\sum c_n(x-a)^n$, one of the following three statements holds:

- (i) There exists a positive real number R such that the series converges absolutely for all x with $|x-a| < R$ but diverges for all x with $|x-a| > R$. The series may or may not converge for x with $|x-a| = R$.
- (ii) The series converges absolutely for all x ($R = \infty$).
- (iii) The series converges at $x = a$ and diverges elsewhere ($R = 0$).

Remark

- The radius of convergence may be found by the Ratio Test (or the Root test).
- If R is the radius of convergence for $\sum c_n(x-a)^n$, where $R \neq 0$ and $R \neq \infty$, then $\sum c_n(x-a)^n$ may or may not converge at $x = a \pm R$.

e.g. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ cvgs for $x \in (-1, 1]$ (last time).

e.g. $\sum x^n$ cvgs for $x \in (-1, 1)$.

One needs to test $x = a \pm R$ separately.

Operations on Power Series

Addition/Subtraction

This is done in the same way as series of numbers.

Multiplication

Let $\sum a_n(x-c)^n$ and $\sum b_n(x-c)^n$ be power series with radii of convergence R_a and R_b , respectively. Let $R := \min\{R_a, R_b\}$. Then

$$\left(\sum_{n=0}^{\infty} a_n(x-c)^n\right)\left(\sum_{n=0}^{\infty} b_n(x-c)^n\right) = \sum_{n=0}^{\infty} C_n(x-c)^n$$

for all x with $|x-c| < R$, where

$$C_n = a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0 = \sum_{k=0}^n a_k b_{n-k}.$$

Idea:

$(a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \dots)(b_0 + b_1(x-c) + b_2(x-c)^2 + b_3(x-c)^3 + \dots)$

coefficient of $(x-c)$ is $a_1 b_0 + a_0 b_1$

To get $(x-c)^n$, can get it from $a_k(x-c)^k \cdot b_{n-k}(x-c)^{n-k}$, $0 \leq k \leq n$.

Substitution

Substitution can be done within interval of convergence.

e.g., Since $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for all $x \in (-1, 1)$, we have
 $\sum_{n=0}^{\infty} (3x)^n = \frac{1}{1-3x}$ valid for $|3x| < 1$, i.e., $x \in (-\frac{1}{3}, \frac{1}{3})$.

Differentiation and Integration

Theorem (Term-by-Term Differentiation and Integration)

Suppose that $\sum_{n=0}^{\infty} c_n(x-a)^n$ has a radius of convergence R , with $R > 0$. Define f on $(a-R, a+R)$ by

$$f(x) := \sum_{n=0}^{\infty} c_n(x-a)^n.$$

Then on $(a-R, a+R)$, the function f is differentiable and has an antiderivative, with

$$(i) \quad f'(x) = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1}.$$

$$(ii) \quad \int f(x) dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}.$$

• Another way to view it is: if $\sum_{n=0}^{\infty} c_n(x-a)^n$ convs to $f(x)$ on $(a-R, a+R)$, then $\sum_{n=1}^{\infty} n c_n(x-a)^{n-1}$ convs to $f'(x)$ on $(a-R, a+R)$, and $\sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}$ convs to an antiderivative of $f(x)$ on $(a-R, a+R)$.

e.g. $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n = \sum_n (-1)^n x^n$ for $|x| < 1$

$$\Rightarrow \int \frac{1}{1+x} dx = C + \sum_{n=0}^{\infty} \frac{1}{n+1} (-1)^n x^{n+1} \quad \text{for } |x| < 1$$

$$\Rightarrow \ln(1+x) = C + \sum_{n=0}^{\infty} \frac{1}{n+1} (-1)^n x^{n+1} \quad \text{for } |x| < 1$$

$(C = \ln(1+x))$

For $x=0$, $0 = \ln 1 = C + 0 \Rightarrow C = 0$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{1}{n+1} (-1)^n x^{n+1} = \ln(1+x) \quad \text{for } x \in (-1, 1).$$

For an approach using definite integrals, see e.g. 10.7.6.

e.g. Let $f(x) := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$.

• By Ratio Test, $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|^{2n+3}}{2n+3} \cdot \frac{2n+1}{|x|^{2n+1}} = \left(\frac{2n+1}{2n+3} \right) |x|^2 \rightarrow |x|^2$,

so series converges when $|x|^2 < 1 \Rightarrow |x| < 1$
and diverges when $|x|^2 > 1 \Rightarrow |x| > 1$

• f is defined on $(-1, 1)$. What is an elementary function expression?

• $f'(x) = \sum_{n=0}^{\infty} (-1)^n x^{2n} = \sum_{n=0}^{\infty} (x^2)^n = \frac{1}{1+x^2}$, for $|x| < 1$

$$\Rightarrow f(x) = \int f'(x) dx = \int \frac{1}{1+x^2} dx = \arctan x + C$$

• $0 = \overset{\substack{\uparrow \\ \text{Series expression}}}{f(0)} = \arctan 0 + C = C \Rightarrow C = 0$

$$\Rightarrow f(x) = \arctan x, \quad x \in (-1, 1).$$

Remark: After term-by-term differentiation/integration, convergence at endpoints $a \pm R$ is NOT guaranteed; e.g. $\sum \frac{x^n}{n^2}$.

$$\cdot \frac{|x|^{n+1}}{(n+1)^2} \cdot \frac{n^2}{|x|^n} = \left(\frac{n}{n+1}\right)^2 |x| \rightarrow |x| \Rightarrow R=1.$$

$$\cdot \text{At } x=1, \sum_{n=0}^{\infty} \frac{x^n}{n^2} \text{ convs.}$$

$$\cdot \text{But } \sum_{n=1}^{\infty} n \frac{x^{n+1}}{n^2} = \sum_{n=1}^{\infty} \frac{x^{n+1}}{n} \text{ does not converge at } x=1.$$

Taylor Series

Suppose $\sum_n C_n (x-a)^n = f(x)$ on some open interval I . By

$$\text{previous theorem, } \sum_{n=1}^{\infty} n C_n (x-a)^{n-1} = f'(x),$$

$$\sum_{n=2}^{\infty} n(n-1) C_n (x-a)^{n-2} = f''(x), \dots$$

i.e. if f has a power series representation, it must be **infinitely differentiable** on I ($f^{(n)}(x)$ exists for all $n \geq 0$).

Conversely, given an infinitely differentiable function f on I :

Q1: Can f be represented as a power series on I ?

Q2: If so, what is the form of this power series?

We look at Q2 first. Suppose $f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n$.

$$\cdot f(a) = \sum C_n (a-a)^n = C_0.$$

$$\cdot f'(a) = \sum_{n=1}^{\infty} n C_n (a-a)^{n-1} \Rightarrow f'(a) = C_1.$$

$$\cdot f''(a) = \sum_{n=2}^{\infty} n(n-1) C_n (a-a)^{n-2} \Rightarrow f''(a) = 2 \cdot 1 \cdot C_2$$

$$\Rightarrow C_2 = \frac{f''(a)}{2!}$$

$$\cdot f^{(k)}(a) = \sum_{n=k}^{\infty} n(n-1)\dots(n-k+1) C_n (x-a)^{n-k} \Big|_{x=a} = k(k-1)\dots 1 \cdot C_k$$

$$\Rightarrow C_k = \frac{f^{(k)}(a)}{k!}$$

Hence, if $f(x)$ has a power series representation $\sum_{n=0}^{\infty} C_n (x-a)^n$, it must be

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Definition

Let f be a function such that for all $n \in \mathbb{N}$, $f^{(n)}$ exists on some open interval containing a . The Taylor series of f centered at a is the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n. \quad \text{or "Taylor series generated by } f \text{"}$$

The Maclaurin series of f is the Taylor series of f centered at 0:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

Example

- (a) The Maclaurin series of $f(x) := e^x$ is $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
- (b) The Maclaurin series of $f(x) := \cos x$ is $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$.

Partial sums of a Taylor series are **Taylor polynomials**. More generally:

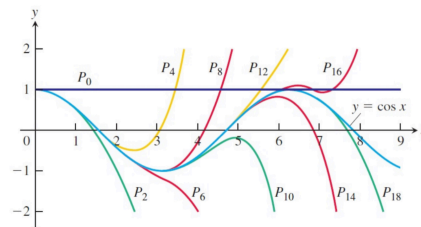
Definition

Let f be a function such that for all $n \in \{0, 1, \dots, N\}$, $f^{(n)}$ exists on some open interval containing a . The **Taylor polynomial of f (of order n) centered at a** is the polynomial

$$P_n(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Remarks:

- For $f(x) = \cos x$ and $a=0$, since $\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots$, we have $P_5(x) = P_4(x) = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$: $P_n(x)$ may have degree $< n$, in general.

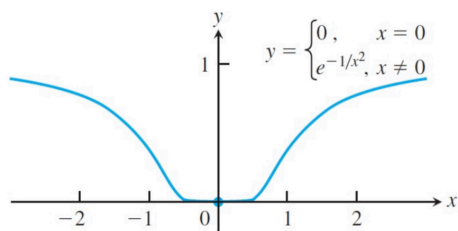


- The Taylor series of f always converges to $f(a)$ at $x=a$:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (a-a)^n = \frac{f^{(0)}(a)}{0!} = f(a).$$

However, in general, $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$ may fail to converge to $f(x)$ for all other values of x .

e.g. $f(x) := \begin{cases} e^{-1/x^2}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$



Can be shown that:

- f is infinitely differentiable
- $f^{(n)}(0) = 0, \forall n \geq 0$.
- So its Taylor series $\equiv 0$.
- But $f(x) \neq 0$ unless $x = 0$.

Details omitted. Search

"Non-analytic smooth function".

Q: (Q1 before) When can f be represented as its Taylor series on an neighbourhood of the center a ?
open interval containing a

Taylor's Theorem

Suppose:

- $f, f', \dots, f^{(n)}$ are continuous on $[a, x]$ (or $[x, a]$);
- $f^{(n+1)}$ exists on (a, x) (or (x, a)).

Then there exists $c \in (a, x)$ (or $c \in (x, a)$) such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

$P_n(x)$

$R_n(x)$: Remainder.

Remarks

- By taking $n=0$, we get $f(x) = f(a) + f'(a)(x-a)$, the M/T, as a special case.
- If for a given $x (\neq a)$, $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$, then

$$f(x) = \lim_{n \rightarrow \infty} f(x) = \lim_{n \rightarrow \infty} P_n(x) + \lim_{n \rightarrow \infty} R_n(x) = \lim_{n \rightarrow \infty} P_n(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Hence, $R_n(x) \rightarrow 0$ gives a sufficient condition for the Taylor series of f to converge to f (at this x)!

In particular, for a given x , if there is a positive constant M such that $|f^{(n+1)}(t)| \leq M$ for all t between a and x , then

$\forall n$

$$|R_n(x)| \leq M \frac{|x-a|^{n+1}}{(n+1)!},$$

so $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$. In this case,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

Example

- Show that the Maclaurin series of $f(x) := e^x$ converges to $f(x)$ for all $x \in \mathbb{R}$. (e.g. 10.9.1)
- Show that the Maclaurin series of $f(x) := \sin x$ converges to $f(x)$ for all $x \in \mathbb{R}$. (e.g. 10.9.2)